

Education	National Taiwan University , Taipei, Taiwan <i>Ph.D., Graduate Institute of Communication Engineering</i> Advisor: Prof. Shao-Hua Sun	Sep. 2019 – Jan. 2025
	National Taiwan University , Taipei, Taiwan <i>B.S. in Electrical Engineering</i>	Sep. 2013 – Jan. 2018
Research Summary	My research centers on generative models and their applications in real-world scenarios. At Spuree, an AI startup focused on 3D animation, I built generative models and video understanding pipelines for animation, and delivered them as cloud APIs. Before that, my Ph.D. research explored generative models for image generation and robot learning.	
Experience	Spuree <i>Member of Technical Staff</i>	May 2025 – Jul. 2026
	• Researched and prototyped generative models for 3D animation, including character motion in-betweening and scene generation. • Built video understanding pipelines for video-driven animation, combining object detection, motion tracking, and depth estimation. • Integrated research prototypes into Spuree’s architecture, delivering animation tools as cloud APIs for Unreal Engine and custom animation platforms.	
	Sony AI <i>Research Intern, Deep Generative Model Team</i> Mentor: Dr. Chieh-Hsin Lai	Jan. 2024 – Aug. 2024
	• Investigated human-feedback-efficient reinforcement learning for fine-tuning text-to-image diffusion models. • Enhanced reasoning, distortion correction, and content safety of the generated images through online human feedback.	
Publications	Inventec AI <i>Research Intern</i> Mentor: Dr. Trista Pei-Chun Chen	May 2021 – Aug. 2022
	• Developed a domain-generalized framework for textured surfaces anomaly-detection. • Performed patch-level anomaly localization to identify anomalous regions.	
	[1] Shang-Fu Chen* , Co Yong*, and Shao-Hua Sun. <i>Restoring Noisy Demonstration for Imitation Learning With Diffusion Models</i> . <i>IEEE Transactions on Neural Networks and Learning Systems (TNNLS)</i> , 2025.	
	[2] Ayano Hiranaka*, Shang-Fu Chen* , Chieh-Hsin Lai*, Dongjun Kim, Naoki Murata, Takashi Shibuya, Wei-Hsiang Liao, Shao-Hua Sun, and Yuki Mitsufuji. <i>HERO: Human-Feedback-Efficient Reinforcement Learning for Online Diffusion Model Fine-tuning</i> . <i>International Conference on Learning Representations (ICLR)</i> , 2025.	
Publications	[3] Shang-Fu Chen* , Hsiang-Chun Wang*, Ming-Hao Hsu, Chun-Mao Lai, and Shao-Hua Sun. <i>Diffusion Model-Augmented Behavioral Cloning</i> . <i>International Conference on Machine Learning (ICML)</i> , 2024.	
	[4] Shang-Fu Chen , Yu-Min Liu, Chia-Ching Lin, Trista Pei-Chun Chen, and Yu-	

Chiang Frank Wang. *Domain-Generalized Textured Surface Anomaly Detection*. *IEEE International Conference on Multimedia and Expo (ICME)*, 2022.

- [5] Zih-Ching Chen^{*}, Lin-Hsi Tsao^{*}, Chin-Lun Fu^{*}, **Shang-Fu Chen**, and Yu-Chiang Frank Wang. *Learning Facial Liveness Representation for Domain Generalized Face Anti-Spoofing*. *IEEE International Conference on Multimedia and Expo (ICME)*, 2022.
- [6] **Shang-Fu Chen**, Jia-Wei Yan, Ya-Fan Su, and Yu-Chiang Frank Wang. *Representation Decomposition for Image Manipulation and Beyond*. *IEEE International Conference on Image Processing (ICIP)*, 2021.
- [7] Yen-Ting Liu, Yu-Jhe Li, Fu-En Yang, **Shang-Fu Chen**, and Yu-Chiang Frank Wang. *Learning Hierarchical Self-Attention for Video Summarization*. *IEEE International Conference on Image Processing (ICIP)*, 2019.
- [8] **Shang-Fu Chen**^{*}, Yi-Chen Chen^{*}, Chih-Kuan Yeh, and Yu-Chiang Frank Wang. *Order-Free RNN With Visual Attention for Multi-Label Classification*. *AAAI Conference on Artificial Intelligence (AAAI)*, 2018.

^{*}Equal contribution.

Awards	Best Ph.D. Thesis Award	2025
	Graduate Institute of Communication Engineering, National Taiwan University <i>Adaptive Machine Learning Pipelines for Computer Vision and Robotic Applications</i>	
Teaching	Reinforcement Learning	Fall 2023
	Graduate Teaching Assistant; Instructor: Prof. Shao-Hua Sun	
	Deep Learning for Computer Vision	Spring 2019
	Graduate Teaching Assistant; Instructor: Prof. Yu-Chiang Frank Wang	
	The Design and Analysis of Algorithms	Fall 2017
Patents	Teaching Assistant; Instructor: Prof. Yu-Chiang Frank Wang	
	Machine Learning	Spring 2017
	Graduate Teaching Assistant; Instructor: Prof. Hung-Yi Lee	
	Taiwan Invention Patent No. I779784	
	<i>Feature Analysis System, Method and Computer Readable Medium Thereof</i> Ya-Fan Su, Hui-Liang Yu, Heng-Sung Liu, Shang-Fu Chen , and Yu-Chiang Frank Wang	
	Taiwan Invention Patent No. I807851	
	<i>A Feature Disentanglement System, Method and Computer-Readable Medium Thereof for Domain Generalized Face Anti-Spoofing</i> Yu-Hsin Cheng, Ya-Fan Su, Heng-Sung Liu, Shang-Fu Chen , and Yu-Chiang Frank Wang	